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10 CFR 50.73

GNRO2024-00037

November 22, 2024

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1 Licensee Event Report 2024-004-00,
High Pressure Core Spray Over Frequency Relay Trip

Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
Renewed License No. NPF-29

Attached is Licensee Event Report (LER) 2024-004-00, High Pressure Core Spray Over Frequency Relay Trip. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(v)(D), any event or condition that could have prevented the fulfillment of the safety function of systems that are needed to mitigate the consequences of an accident.

This letter contains no new Regulatory Commitments. Should you have any questions concerning the content of this letter, please contact me at 802-380-5124.

Sincerely,

A handwritten signature in blue ink, appearing to read "JAH", with a stylized flourish at the end.

JAH/saw

Attachments: Licensee Event Report 2024-004-00

cc: NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

U.S Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Attachment
Licensee Event Report 2024-004-00



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Grand Gulf Nuclear Station, Unit 1

☒ 050
☐ 052

2. Docket Number

00416

3. Page

1 of 2

4. Title

High Pressure Core Spray Over Frequency Relay Trip

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	☐ 050 Docket Number
09	24	2024	2024	004	00	11	22	2024	N/A	☐ 052 Docket Number

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)(i)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ Other (Specify here, in abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact
Jeff Hard, Regulatory Assurance ManagerPhone Number (Include Area Code)
802-380-5124

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
B	BG	RLY		N					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On September 24, 2024, Grand Gulf Nuclear Station, Unit 1 was operating at 100 percent power with the High Pressure Core Spray (HPCS) offline for testing and all safety systems operable. At 22:04 CT, during planned surveillance testing of HPCS and the Division III diesel generator, the HPCS pump breaker unexpectedly tripped after the HPCS diesel generator started and powered the safety bus. The breaker performed its motor protection function and tripped due to an over-frequency indication. This resulted in a potential loss of safety function. Troubleshooting was performed, and the HPCS over-frequency relay was replaced. The system was returned to service. The cause of the trip was determined to be the failure of the over-frequency relay. The over-frequency relay was determined to be non-essential, and actions were created to remove the relay from the system. During the testing of the HPCS system, alternate ECCS systems were available to provide core cooling and injection to the reactor; therefore, there were no actual safety consequences to the event.

This report is made in accordance with 10 CFR 50.73a(2)(v)(D) for an event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Grand Gulf Nuclear Station, Unit 1	☒ 050	00416	YEAR	SEQUENTIAL NUMBER	REV NO.
	☐ 052		2024	004	00

NARRATIVE**Plant Conditions:**

At the time of the event, Grand Gulf Nuclear Station, Unit 1 was in Mode 1 at 100% power. The station was conducting planned surveillance testing on the High Pressure Core Spray (HPCS) Division III Emergency Diesel Generator (EDG).

Description of Event:

On September 24, 2024, while performing a HPCS loss of offsite power surveillance, the Division III EDG started and loaded the safety bus as expected, but the HPCS pump breaker immediately tripped which resulted in a potential loss of safety function.

Reportability:

This report is made in accordance with 10 CFR 50.73a(2)(v)(D) for an event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. HPCS is required to be able to inject water during a loss of cooling accident.

Cause(s):

The direct cause of the event was a failure of the HPCS over-frequency relay. The HPCS original system design included an over-frequency relay that is not required. This feature has existed since construction, and its need was not evaluated after HPCS was built without the Level-8 injection valve isolation bypass signal, per the GE system design criteria.

Corrective Actions:**Completed Actions:**

- Electrical Maintenance replaced the HPCS over-frequency relay.
- A Failure Modes Analysis and a Root Cause Evaluation were performed to identify and correct the causes of the event.

Planned Actions:

- Revise current preventative maintenance to include capacitor checks based on EPRI guidance.
- Revise over-frequency calibration procedure to state that all installation or reinstallation of the over frequency relay is required to be with a matched capacitor.
- Remove the over-frequency relay protection (17AC-181) from the HPCS motor protection circuit.

Safety Significance:

There were no actual safety consequences associated with this condition. The HPCS system functions as a part of the Emergency Core Cooling System (ECCS). It functions to provide and maintain an adequate coolant inventory inside the reactor vessel to limit the fuel cladding temperatures in the event of breaks in the reactor coolant pressure boundary. The system is initiated by either high pressure in the drywell or low water level in the reactor vessel. During the testing of the HPCS system, alternate ECCS systems were available to provide core cooling and injection to the reactor.

Previous Similar Events:

LER 24-001-00 High Pressure Core Spray Inoperable due to Minimum Flow Valve Failure to Close